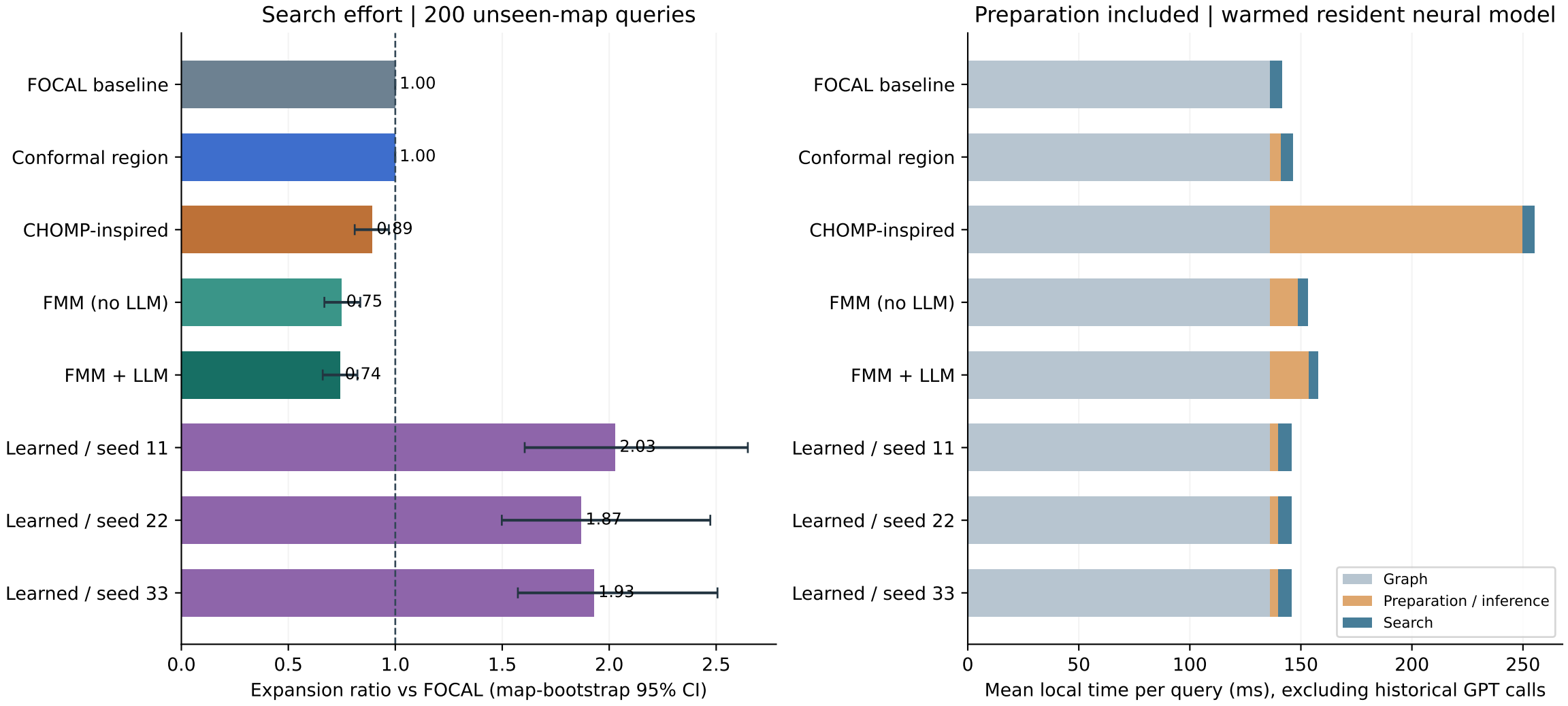
Search effort | 200 unseen-map queries
FOCAL baseline
Conformal region
CHOMP-inspired
FMM (no LLM)
FMM + LLM
Learned / seed 11
Learned / seed 22
Learned / seed 33
1.00
1.00
0.89
0.75
0.74
2.03
1.87
1.93
0.0
0.5
1.0
1.5
2.0
2.5
Expansion ratio vs FOCAL (map-bootstrap 95% CI)
Preparation included | warmed resident neural model
0
50
100
150
200
250
Mean local time per query (ms), excluding historical GPT calls
Graph
Preparation / inference
Search